AI Analyst

System & Modeling Reference

MZQA AI Investment Committee

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Abstract

AI Analyst is a multi-agent equity-research application: a LangGraph "investment committee" of LLM analysts deliberating over a deterministic valuation engine, a value/sentiment screener, and a quantitative desk that adds machine-learned return and risk forecasting and portfolio optimization built on Microsoft `qlib`. This document describes the system and gives the *complete mathematical specification* of every modeling approach it uses: the gradient-boosted alpha model, the factor-structured risk models, the portfolio optimizers, the walk-forward out-of-sample backtest with Fama–French benchmarking, the macro-regime models (HMM / VAE), the rolling factor regression, and the deterministic valuation and scoring models.

Contents

Part I System

1 Overview and architecture

AI Analyst turns a question — *"what is this stock worth, and what should I expect from it?"* — into a structured, evidence-anchored answer. It has three qualitative entry modes (single-stock committee, group/industry relative-value verdict, and an idea scanner) and a quantitative *Quant* desk.

Layer	Technology
Backend	FastAPI (async), Uvicorn, port 8027; blocking work on worker threads
Committee engine	LangGraph `StateGraph`; custom LLM-calling nodes
Quant layer	`qlib` (LightGBM, scikit-learn, cvxpy) — lean in-process embed
Frontend	Next.js 14 / React 18 / Tailwind, port 3027 (REST only)
Database	read-only PostgreSQL `xbrl_sec` (schema `sec`)
LLMs	5 providers (DeepSeek, OpenAI, Anthropic, Moonshot, Gemini), per request

Three backend packages: `ai_analyst/committee` (the tribunal graph + valuation + evidence), `api` (REST routers + the `api/quant` modeling layer), and `xbrl_sec` (data access, the macro-cycle models, and the LLM client).

2 Data warehouse

A pre-built, read-only warehouse is the single source of truth; the app never writes to it. Key tables: prices (`fact_prices_{us,jp,etf}`), fundamentals (`fact_metrics_{us,jp}`, `fact_fundamentals_std_{us,jp}`), factors (`fact_fama_french`, `fact_factor_loadings`, `fact_factor_reg_meta`), macro regime (`fact_cycle_state_month` `fact_equity_factor_ic_regime`), text/sentiment (`fact_mda_sections_{us,jp}`), and 13F ownership. Fundamentals are aligned point-in-time: a value with fiscal period end p becomes usable only at monthend($p + 90$ days), so a signal never uses data that was not yet public.

3 The investment committee

A LangGraph state machine over an `InvestmentCommitteeState`. It splits into a *prepare* phase (deterministic gate → financial-analysis engine → evidence nodes: news/macro, 13F, data-quality agent, *qlib quant signals*) and a *debate* phase (Advocate, Challenger, Auditor and sector specialists → Lead → memo). The prepare phase is provider-independent, so several LLM providers can debate the same evidence concurrently. Specialists include a *Quantitative Factor Analyst* that interprets the quant-desk signals. Output: a probability-weighted fair value, a triangulation range, scenarios, an evidence bundle, and a written memo — a valuation with its reasoning, not a single buy/sell label.

4 Screener and idea generation

The screener filters the US/JP/INTL universe on fundamentals (deterministic SQL) and offers a natural-language screen (LLM → filters). The *value + sentiment* scanner produces a 0–100 "interest score" (§14.1); the group ranking produces a deterministic relative-value composite (§14.2).

5 The quant desk

A lean, in-process integration of `qlib`: a cross-sectional alpha model (return prediction, §7), a factor-structured risk model (§8), dual portfolio optimization (native *and* qlib backends, §9), and a walk-forward out-of-sample backtest with Fama–French benchmarking (§10). Exposed via a `/api/quant/*` router and a *Quant* UI tab, and fed into the committee as an evidence node.

Part II Mathematical modeling reference

6 Notation

$\mathbf{w} \in \mathbb{R}^N$ portfolio weights; $\boldsymbol{\mu} \in \mathbb{R}^N$ expected (annualized) returns; $\Sigma \in \mathbb{R}^{N\times N}$ return covariance; r_f risk-free rate; $\mathbf{1}$ the ones vector; z a cross-sectional z-score. Time subscripts t index months unless stated; annualization uses 252 trading days (daily) or 12 (monthly).

7 Cross-sectional alpha model (return prediction)

`api/quant/qlib_alpha.py`, built on `qlib`'s `LGBModel` (LightGBM).

7.1 Feature/label panel

Monthly, cross-sectional. For each name i and month t, raw fundamental factors $x_{i,t}^{(j)}$ (value / quality / growth / market-factor families) are *cross-sectionally* standardized and winsorized,

$$z_{i,t}^{(j)} \;=\; \mathrm{clip}\left(\frac{x_{i,t}^{(j)} - \bar{\mu}_t^{(j)}}{\hat{\sigma}_t^{(j)}},\, -3,\, 3\right), \qquad \bar{\mu}_t^{(j)} = \frac{1}{N_t}\sum_i x_{i,t}^{(j)},\;\; \hat{\sigma}_t^{(j)} = \mathrm{std}_i\, x_{i,t}^{(j)}, \tag{1}$$

so features are comparable across names/dates and robust to scale. The label is the forward one-month (or three-month) total return $y_{i,t} = r_{i,t\to t+1}$.

7.2 Gradient-boosted decision trees

The model is an additive ensemble of regression trees,

$$F_M(\mathbf{z}) = \sum_{m=1}^{M} \eta\, h_m(\mathbf{z}), \tag{2}$$

with learning rate η. For the squared-error objective $\mathcal{L} = \sum_i \frac{1}{2}\big(y_i - F(\mathbf{z}_i)\big)^2$, boosting stage m fits h_m to the negative gradient (the residual) $g_i = -\partial\mathcal{L}/\partial F = y_i - F_{m-1}(\mathbf{z}_i)$, with Hessian $h_i = 1$. LightGBM grows trees *leaf-wise* using a histogram approximation; a split of node with instance set I into I_L, I_R has gain

$$\mathrm{Gain} = \tfrac{1}{2}\left[\frac{\big(\sum_{i\in I_L} g_i\big)^2}{\sum_{i\in I_L} h_i + \lambda} + \frac{\big(\sum_{i\in I_R} g_i\big)^2}{\sum_{i\in I_R} h_i + \lambda} - \frac{\big(\sum_{i\in I} g_i\big)^2}{\sum_{i\in I} h_i + \lambda}\right] - \gamma, \tag{3}$$

where λ is the L_2 leaf regularizer and γ the split penalty. Trees capture non-linearities and factor interactions a linear model cannot. Training uses regularization λ_1, λ_2, `num_leaves`, `max_depth`, and feature/row subsampling, with early stopping on a validation split. (Our `AlphaLGB` subclass omits `qlib`'s recorder logging so training needs no `qlib.init`/mlflow.)

7.3 Evaluation — information coefficient

Per month t, with prediction $\hat{y}_{i,t}$,

$$\mathrm{IC}_t = \mathrm{Corr}\,\big(\hat{y}_{\cdot,t}, y_{\cdot,t}\big)\ \text{(Pearson)}, \quad \mathrm{RankIC}_t = \rho_{\mathrm{Spearman}}\big(\hat{y}_{\cdot,t}, y_{\cdot,t}\big), \tag{4}$$

and the information ratio of the signal $\mathrm{ICIR} = \overline{\mathrm{IC}}/\,\mathrm{std}(\mathrm{IC})$. The shipped US/JP models reach test-segment $\overline{\mathrm{RankIC}} \approx 0.08$–$0.14$.

8 Factor-structured risk model

`api/quant/qlib_risk.py` wraps `qlib`'s `StructuredCovEstimator`.

Returns are assumed to be driven by K latent factors,

$$X = B\,F^{\mathsf{T}} + U, \qquad \mathrm{Cov}(X) = F\;\mathrm{Cov}(B)\,F^{\mathsf{T}} + \mathrm{diag}\big(\mathrm{Var}(U)\big), \tag{5}$$

where $X \in \mathbb{R}^{T\times N}$ is the return panel, $F \in \mathbb{R}^{N\times K}$ are factor *exposures* (the principal components, from PCA; Factor Analysis is also supported), $B \in \mathbb{R}^{T\times K}$ the factor *returns* (component scores), and U the idiosyncratic residual. Writing $\Sigma_B = \mathrm{Cov}(B)$ and $\mathbf{d} = \mathrm{Var}(U)$ (diagonal),

$$\boxed{\Sigma \;=\; F\,\Sigma_B\,F^{\mathsf{T}} + \mathrm{diag}(\mathbf{d})} \tag{6}$$

which is a low-rank-plus-diagonal shrinkage of a noisy $N \times N$ sample covariance, far more stable for optimization. We estimate on raw returns (`scale_return=False`) and annualize $\Sigma_B, \mathbf{d}$ by $\times 252$. Forward volatility per name is $\sigma_i = \sqrt{\Sigma_{ii}}$; the decomposition $(F, \Sigma_B, \mathbf{d})$ feeds the enhanced-indexing optimizer.

8.1 Ledoit–Wolf shrinkage (alternative)

`api/quant/risk.py` also offers analytic shrinkage toward a constant-correlation target. With sample covariance S, s_{ii} variances, and mean pairwise correlation $\bar{r} = \frac{2}{N(N-1)}\sum_{i<j} s_{ij}/\sqrt{s_{ii}s_{jj}}$, the target is $F_{ii} = s_{ii}$, $F_{ij} = \bar{r}\sqrt{s_{ii}s_{jj}}$, and

$$\Sigma^\star = \delta F + (1-\delta)S, \qquad \delta = \mathrm{clip}\Big(\tfrac{\pi/\gamma}{T}, 0, 1\Big), \tag{7}$$

where $\gamma = \|S - F\|_F^2$ measures target misspecification and $\pi = \sum_{i,j}\big[\frac{1}{T}\sum_t (x_{ti}x_{tj} - s_{ij})^2\big]$ is the summed asymptotic variance of S. Result annualized $\times 252$.

8.2 Fama–French structured covariance (alternative)

$\Sigma = B\,F_{\mathrm{ann}}\,B^{\mathsf{T}} + D$, with B the per-name FF loadings (`fact_factor_loadings`, §11), $F_{\mathrm{ann}} = 252\;\mathrm{Cov}(\text{factor returns})$, and $D = \mathrm{diag}(\text{residual\_vol}_i^2)$ from the regression — a *fundamental* (named-factor) analogue of the statistical PCA covariance above.

9 Portfolio optimization (dual backend)

`api/quant/qlib_optimize.py` dispatches to a *native* optimizer or any `qlib` optimizer at runtime; both are fed the same $\boldsymbol{\mu}, \Sigma$.

9.1 Native optimizer

`api/quant/optimize.py` maximizes a multi-term mean–variance utility,

$$\max_{\mathbf{w}} \;\; \mathbf{w}^{\mathsf{T}}\boldsymbol{\mu} - \tfrac{1}{2}\lambda\,\mathbf{w}^{\mathsf{T}}\Sigma\mathbf{w} - \gamma_f\big\|S \odot a \odot (B^{\mathsf{T}}\mathbf{w} - \mathbf{t})\big\|_2^2 - \gamma_{\mathrm{to}}\|\mathbf{w} - \mathbf{w}_0\|_1 - \gamma_c\|\mathbf{w}\|_2^2, \tag{8}$$

subject to full investment $\mathbf{1}^{\mathsf{T}}\mathbf{w} = 1$, per-name bounds $l_i \le w_i \le u_i$, gross cap $\sum_i |w_i| \le G$, sector caps $\mathbf{a}_s^{\mathsf{T}}\mathbf{w} \le c_s$, an annualized volatility cap $\mathbf{w}^{\mathsf{T}}\Sigma\mathbf{w} \le \sigma_{\max}^2$, and factor caps/ranges $|\,\mathbf{b}_j^{\mathsf{T}}\mathbf{w}\,| \le \mathrm{cap}_j$. Here λ is risk aversion, γ_f penalizes deviation of factor exposures $B^{\mathsf{T}}\mathbf{w}$ from targets $\mathbf{t}$ (scaled by S, active mask a), γ_{to} penalizes turnover vs. $\mathbf{w}_0$, and γ_c penalizes concentration. Solved by **SLSQP** with the analytic gradient

$$\nabla = -\boldsymbol{\mu} + \lambda\Sigma\mathbf{w} + 2\gamma_f B\big(S^2 \odot a \odot (B^{\mathsf{T}}\mathbf{w} - \mathbf{t})\big) + \gamma_{\mathrm{to}}\,\mathrm{sign}(\mathbf{w} - \mathbf{w}_0) + 2\gamma_c\mathbf{w}. \tag{9}$$

When a cardinality constraint $\|\mathbf{w}\|_0 \le K$ is set it becomes a **mixed-integer program** with binaries $z_i \in \{0,1\}$, $w_i \le z_i$, $\sum_i z_i \le K$, solved via cvxpy (HiGHS/SCIP/CBC). It also sweeps the efficient frontier.

9.2 qlib optimizers

All are long-only and fully invested ($\mathbf{1}^\mathsf{T}\mathbf{w} = 1,\ \mathbf{w} \geq 0$):

Backend	Objective
GMV (min variance)	$\min_{\mathbf{w}}\ \mathbf{w}^\mathsf{T}\Sigma\mathbf{w}$
MVO (mean–variance)	$\max_{\mathbf{w}}\ \lambda\,\tilde{r}^\mathsf{T}\mathbf{w} - \mathbf{w}^\mathsf{T}\Sigma\mathbf{w}$ ($\tilde{r}$ scaled to Σ's vol)
RP (risk parity)	choose $\mathbf{w}$ s.t. $w_i(\Sigma\mathbf{w})_i = w_j(\Sigma\mathbf{w})_j\ \forall i, j$
Inverse-vol	$w_i = \dfrac{1/\sigma_i}{\sum_j 1/\sigma_j}, \quad \sigma_i = \sqrt{\Sigma_{ii}}$

Risk parity equalizes the risk contributions $\mathrm{RC}_i = w_i(\Sigma\mathbf{w})_i$ (with $\sum_i \mathrm{RC}_i = \mathbf{w}^\mathsf{T}\Sigma\mathbf{w}$). GMV/MVO/RP are solved with `scipy.optimize`.

Enhanced indexing (benchmark-relative). With benchmark weights $\mathbf{w}_b$, deviation $\mathbf{d} = \mathbf{w} - \mathbf{w}_b$ and factor deviation $\mathbf{v} = \mathbf{d}^\mathsf{T}F$,

$$\max_{\mathbf{w}}\ \mathbf{d}^\mathsf{T}\mathbf{r} - \lambda\big(\mathbf{v}^\mathsf{T}\Sigma_B\mathbf{v} + \mathbf{d}^\mathsf{T}\mathrm{diag}(\mathbf{d})\,\mathbf{d}_u\big) \quad \text{s.t.} \quad \mathbf{w} \geq 0,\ \mathbf{1}^\mathsf{T}\mathbf{w} = 1,\ \|\mathbf{w} - \mathbf{w}_0\|_1 \leq \delta,\ |\mathbf{d}| \leq b_{\mathrm{dev}},\ |\mathbf{v}| \leq f_{\mathrm{dev}}, \tag{10}$$

i.e. maximize active return minus active (tracking-error) risk with bounded factor and name deviations. Convex QP solved by cvxpy/ECOS; $\mathbf{d}_u = \mathrm{Var}(U)$ is the specific variance.

Given a solution $\mathbf{w}^\star$ the desk reports $\mathbb{E}[r]_{\mathrm{ann}} = \mathbf{w}^{\star\mathsf{T}}\boldsymbol{\mu}$, $\sigma_{\mathrm{ann}} = \sqrt{\mathbf{w}^{\star\mathsf{T}}\Sigma\,\mathbf{w}^\star}$, and Sharpe $= (\mathbb{E}[r]_{\mathrm{ann}} - r_f)/\sigma_{\mathrm{ann}}$.

10 Walk-forward out-of-sample backtest

`api/quant/qlib_backtest.py`. `qlib`'s exchange backtest needs a `.bin` data provider the lean embed lacks, so we run a cross-sectional signal backtest that is *out-of-sample by construction.*

10.1 Walk-forward predictions

Let $t_0 < t_1 < \cdots < t_T$ be the panel months, restricted to an investable universe (marketcap $\geq$ \$2B), with each month's forward returns winsorized to the 1st/99th cross-sectional percentile. For each test month t_i ($i \geq m_{\min}$, refit every q months) a booster g_{t_i} is trained on *all prior* months $\{t_j : t_j < t_i\}$ and produces $\hat{\alpha}_{k,t_i} = g_{t_i}(\mathbf{z}_{k,t_i})$. No month is tested on data it trained on.

10.2 Portfolio construction

Ranking names by $\hat{\alpha}$, the equal-weight top-k book realizes

$$r_t^{\mathrm{L}} = \frac{1}{k}\sum_{i\in\text{top-}k} r_{i,t}, \qquad r_t^{\mathrm{LS}} = \frac{1}{k}\sum_{i\in\text{top-}k} r_{i,t} - \frac{1}{k}\sum_{i\in\text{bot-}k} r_{i,t}, \tag{11}$$

for long-only and long/short respectively. (The expensive walk-forward predictions are cached, so changing k or long/short re-ranks instantly.)

10.3 Performance and risk metrics

With monthly return series $\{r_t\}_{t=1}^{n}$, equity $E_t = \prod_{s\le t}(1+r_s)$, monthly risk-free $r_{f,t}$:

$$R_{\text{ann}} = \Big(\prod_t (1+r_t)\Big)^{12/n} - 1, \qquad \sigma_{\text{ann}} = \operatorname{std}(r_t)\sqrt{12}, \tag{12}$$

$$\text{Sharpe} = \frac{R_{\text{ann}} - R^f_{\text{ann}}}{\sigma_{\text{ann}}}, \qquad \text{Sortino} = \frac{R_{\text{ann}} - R^f_{\text{ann}}}{\operatorname{std}(r_t : r_t < 0)\sqrt{12}}, \tag{13}$$

$$\text{MaxDD} = \min_t \Big(\tfrac{E_t}{\max_{s\le t} E_s} - 1\Big), \qquad \text{Hit} = \tfrac{1}{n}\textstyle\sum_t \mathbf{1}[r_t > 0]. \tag{14}$$

10.4 Fama–French benchmarking

The benchmark is the FF *market* total return $r^m_t = (\text{Mkt-RF})_t + (\text{RF})_t$, compounded from daily `fact_fama_french` values to monthly and aligned to $\{r_t\}$. Relative to it,

$$\text{excess} = R_{\text{ann}} - R^m_{\text{ann}}, \quad \text{TE} = \operatorname{std}(r_t - r^m_t)\sqrt{12}, \quad \text{IR} = \frac{\text{excess}}{\text{TE}}, \quad \beta = \frac{\operatorname{Cov}(r, r^m)}{\operatorname{Var}(r^m)}. \tag{15}$$

10.5 Fama–French 5-factor + momentum regression

To separate genuine alpha from factor exposure we regress the strategy's excess return on the FF factors,

$$r_t - r_{f,t} = \alpha + \beta_{\text{Mkt}} f^{\text{Mkt}}_t + \beta_{\text{SMB}} f^{\text{SMB}}_t + \beta_{\text{HML}} f^{\text{HML}}_t + \beta_{\text{RMW}} f^{\text{RMW}}_t + \beta_{\text{CMA}} f^{\text{CMA}}_t + \beta_{\text{Mom}} f^{\text{Mom}}_t + \varepsilon_t. \tag{16}$$

OLS via least squares on the design $A = [\mathbf{1} \;\; X]$: $\hat\theta = (A^\top A)^{-1}A^\top y$, residual variance $\hat s^2 = \|\varepsilon\|^2/(n-K-1)$, coefficient covariance $\hat s^2(A^\top A)^{-1}$, so the alpha t-statistic is $t_\alpha = \hat\alpha/\operatorname{se}(\hat\alpha)$, annualized alpha $\alpha_{\text{ann}} = (1+\hat\alpha)^{12} - 1$, and $R^2 = 1 - \|\varepsilon\|^2/\sum_t (y_t - \bar y)^2$. In a proper walk-forward the shipped US signal shows $\overline{\text{RankIC}} \approx 0.01$ out-of-sample and $t_\alpha \approx 1.2$ (not significant): weak once you stop testing on training data — the honest result.

11 Rolling Fama–French factor regression

`xbrl_sec/sec/sources/factor_model.py`. Per ticker, over a rolling 252-day window, excess returns are regressed on the FF factors (FF3/FF4/FF5/FF6),

$$r_{i,t} - r_{f,t} = \alpha_i + \sum_k \beta_{i,k}\, f_{k,t} + \varepsilon_{i,t}. \tag{17}$$

Estimation uses the **Huber** robust loss (down-weighting outlier days),

$$\rho_\delta(e) = \begin{cases} \frac{1}{2}e^2 & |e| \le \delta \\ \delta\big(|e| - \frac{1}{2}\delta\big) & |e| > \delta, \end{cases} \tag{18}$$

with **Newey–West** HAC standard errors robust to autocorrelation/heteroskedasticity: $\widehat{\operatorname{Var}}(\hat\beta) = (X^\top X)^{-1}\hat S(X^\top X)^{-1}$ with

$$\hat S = \sum_{\ell=0}^{L} w_\ell \sum_t \big(x_t\varepsilon_t\big)\big(x_{t-\ell}\varepsilon_{t-\ell}\big)^\top, \qquad w_\ell = 1 - \frac{\ell}{L+1} \text{ (Bartlett)}. \tag{19}$$

Stored diagnostics: adj $R^2 = 1 - (1-R^2)\frac{n-1}{n-k}$, residual volatility $\hat\sigma_\varepsilon\sqrt{252}$ (the idiosyncratic D of §8.2), Durbin–Watson, condition number, F-statistic. These loadings feed the WACC (§13.2) and the FF structured covariance.

12 Macro-cycle regime models

`xbrl_sec/sec/cycle/`. A PCA/dynamic-factor baseline first compresses a multimodal feature set (macro, market, fundamental) into latent cycle factors $\mathbf{x}_t$ (`fact_cycle_state_monthly.latent_cycle`). Two regime models run on top.

12.1 Gaussian Hidden Markov Model

Latent states $s_t \in \{1, \dots, K\}$ ($K = 4$) with Gaussian emissions and a Markov transition,

$$\mathbf{x}_t \mid s_t{=}k \sim \mathcal{N}(\boldsymbol{\mu}_k, \Sigma_k), \qquad A_{ij} = P(s_t{=}j \mid s_{t-1}{=}i), \qquad \pi_k = P(s_1{=}k). \tag{20}$$

Parameters $\{\pi, A, \boldsymbol{\mu}_k, \Sigma_k\}$ are fit by Baum–Welch (EM). With forward $a_t(k) = P(\mathbf{x}_{1:t}, s_t{=}k)$ and backward $b_t(k) = P(\mathbf{x}_{t+1:T} \mid s_t{=}k)$, the smoothed posterior is

$$\gamma_t(k) = P(s_t{=}k \mid \mathbf{x}_{1:T}) = \frac{a_t(k)\, b_t(k)}{\sum_{k'} a_t(k')\, b_t(k')}, \tag{21}$$

the confidence is $\max_k \gamma_t(k)$, and the MAP state path comes from Viterbi. States are mapped to business-cycle phases (early/mid expansion, slowdown, contraction, recovery). If `hmmlearn` is unavailable a deterministic k-means-style assignment with empirical transitions is used.

12.2 Temporal multimodal VAE

An optional variational autoencoder: encoder $q_\phi(\mathbf{z} \mid \mathbf{x}) = \mathcal{N}\big(\boldsymbol{\mu}_\phi(\mathbf{x}), \operatorname{diag} \boldsymbol{\sigma}^2_\phi(\mathbf{x})\big)$, decoder $p_\theta(\mathbf{x} \mid \mathbf{z})$, prior $p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, I)$, trained by maximizing the evidence lower bound

$$\mathcal{L}(\theta, \phi) = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\big[\log p_\theta(\mathbf{x} \mid \mathbf{z})\big] - \mathrm{KL}\big(q_\phi(\mathbf{z} \mid \mathbf{x}) \,\|\, p(\mathbf{z})\big), \tag{22}$$

with the reparameterization $\mathbf{z} = \boldsymbol{\mu}_\phi + \boldsymbol{\sigma}_\phi \odot \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, I)$. The latent dimensions are split into interpretable growth / inflation / rates / credit / market / fundamentals factors; the smoothed reconstruction-error stress percentile is thresholded into cycle phases. A PCA surrogate is used when PyTorch is absent.

12.3 Regime-conditioned factor IC

`cycle/ic.py` links the regime to forecasting. For each fundamental factor m, conditioned on the regime,

$$\mathrm{IC}_{\text{regime}}(m) = \rho_{\text{Spearman}}\big(x_{m,i},\, r_i^{\text{fwd}} \;\big|\; \text{regime}\big), \tag{23}$$

with a probability-weighted variant that weights each observation by the regime posterior $\gamma_t(k)$ (soft assignment). Written to `fact_equity_factor_ic_regime`, it says which factors have actually predicted returns in conditions like today's, and drives the scanner's factor-family weights (§14.1).

13 Valuation models

`ai_analyst/`. Deterministic, on a 7-year explicit horizon.

13.1 Discounted cash flow

For each forecast year $i = 1, \dots, 7$,

$$\mathrm{Rev}_i = \mathrm{Rev}_{i-1}(1 + g_i), \qquad \mathrm{EBIT}_i = \mathrm{Rev}_i \cdot m, \tag{24}$$

$$\mathrm{NOPAT}_i = \mathrm{EBIT}_i - \max(\mathrm{EBIT}_i, 0)\,\tau, \qquad \mathrm{FCF}_i = \mathrm{NOPAT}_i + \mathrm{D\&A}_i - \mathrm{Capex}_i - \Delta\mathrm{NWC}_i, \tag{25}$$

with D&A, Capex, ΔNWC taken as fractions of revenue. Terminal value (Gordon growth) and enterprise/equity value:

$$\mathrm{TV} = \frac{\mathrm{FCF}_7(1+g_\infty)}{\mathrm{WACC}-g_\infty}, \qquad \mathrm{EV} = \sum_{i=1}^{7} \frac{\mathrm{FCF}_i}{(1+\mathrm{WACC})^i} + \frac{\mathrm{TV}}{(1+\mathrm{WACC})^7}, \tag{26}$$

Equity = EV − NetDebt, per share = Equity/shares. A 5×5 sensitivity grid varies WACC $\pm$200bp and $g_\infty \pm 100$bp.

13.2 Cost of capital (WACC)

Cost of equity is built from the FF factor betas (§11) and long-run premia $\Pi_k = 252 \cdot \overline{f_k}$:

$$R_e^{\mathrm{CAPM}} = r_f + \beta_{\mathrm{Mkt}}\,\mathrm{ERP}, \qquad R_e^{\mathrm{FF5}} = r_f + \sum_k \beta_k\,\Pi_k, \tag{27}$$

r_f the 10Y Treasury. Cost of debt $R_d = r_f + \mathrm{spread(coverage)}$ from a Damodaran synthetic-rating table on interest coverage EBIT/Interest, and

$$\mathrm{WACC} = \frac{E}{V} R_e + \frac{D}{V} R_d(1-\tau), \qquad V = E + D, \tag{28}$$

E market cap, D total debt. The headline uses R_e^{CAPM}; the multi-factor R_e^{FF5} is reported alongside. `segment_wacc` shifts β and adds a growth-risk premium for the sum-of-the-parts DCF.

13.3 Reverse DCF, scenarios, triangulation

Reverse DCF solves, by bisection, the flat growth $g^\star$ (or steady margin $m^\star$) with $\mathrm{PerShare}(g^\star) =$ price — what the market is pricing in. *Scenario weighting* combines per-share values V_s (upside/base/downside) with probabilities $\sum_s p_s = 1$ into $\mathrm{FV} = \sum_s p_s V_s$. *Triangulation* builds a football field from the SOTP range (primary), the consolidated DCF scenario range, and peer-multiples-implied values, reporting implied upside FV/price − 1.

14 Scoring and ranking models

14.1 Value + sentiment interest score

`api/quant/../screener_agent.py`. Each component is min–max normalized across the shortlist to $[0, 1]$ (v_{PE} inverted — cheaper is better), plus a normalized alpha v_α from the qlib model:

$$\mathrm{score} = w_\alpha\, v_\alpha + (1-w_\alpha) \sum_k w_k\, v_k, \qquad \mathrm{interest} = 100 \cdot \mathrm{score}, \tag{29}$$

with $w_\alpha \approx 0.40$ (dominant when the model covers the shortlist). The fundamental sub-weights w_k (FCF-yield, P/E, growth, tone, news) start from availability-branched defaults and are re-split between the *value* and *growth* blocks by the current regime's factor IC (§12.3): with family emphases $e_{\mathrm{val}}, e_{\mathrm{grw}}$ (normalized mean |IC|), the fundamental budget is allocated $\propto e_{\mathrm{val}} : e_{\mathrm{grw}}$. With no model or IC it reduces exactly to the legacy fixed-weight composite.

14.2 Group relative-value composite

`ai_analyst/committee/group.py`. For each metric k (P/E, EV/EBITDA, P/B, FCF yield, dividend, revenue growth/CAGR, gross/operating margin), a cross-sectional z-score $z_{k,i} = (x_{k,i} - \bar{x}_k)/\hat{\sigma}_k$ and a signed weight w_k (negative for cheaper-is-better multiples). Loss-making multiples (negative P/E, EV/EBITDA, P/B) are excluded from their metric's population. The composite is

$$\text{composite}_i = \sum_k w_k \, z_{k,i}, \tag{30}$$

ordered descending; terciles map to attractive / fair / expensive. The per-name audit trail ($x \to z \to \times w \to$ contribution) sums exactly to the composite. An optional LLM adds a narrative; the ranking itself is deterministic.

14.3 MD&A management tone

An LLM reads an excerpt of each name's latest MD&A and returns a structured tone in $[-1, 1]$, a positive/neutral/negative guidance label, and risk flags; the tone feeds v_{tone} above.

Part III Reference

15 How the models connect

The *valuation* path (DCF/WACC/SOTP/reverse/triangulation) answers what a business is worth; the *quant* path (alpha $\rightarrow \boldsymbol{\mu}$, structured covariance $\rightarrow \Sigma$, optimizer $\rightarrow \mathbf{w}$, backtest) answers what to expect and how to size it; the *macro-cycle* path (HMM/VAE $\rightarrow$ regime-conditioned IC) conditions the scoring on the regime; and the *committee* fuses the deterministic evidence and the quant signals into a reasoned, auditable verdict.

16 REST API (active surface)

Base `http://127.0.0.1:8027`, under `/api`: `meta`, `screener` (+`/ai`), `screener/agent/value-sentiment`, `ai/committee` (+`/prepare`,`/debate`,`/iterate`), `ai/committee/group`, `sector`, `prices`, `kpis`, `company`, `fx`, `llm`, and the quant desk `quant/{backends,alpha,risk,optimize,backtest}`.

17 Running

Backend (venv Python, from `backend/` so `qlib` resolves): `uvicorn api.main:app --port 8027`; frontend `npm run dev` (port 3027); or `start_committee.bat`. `qlib` is built from the bundled clone (no PyPI wheel for Python 3.13) and installed *non-editable* so `import qlib` resolves from any directory. The quant desk pre-warms the alpha cross-section and the walk-forward backtest at startup, so the first user request is served from cache.